Supplementary Material

TrustAero: Agentic Query Planning and Auditable Execution for Governed Data Processing

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A. Experimental Environment and Reporting Conventions

The supplementary experiments use the same implementation and measurement conventions as the main paper. Runtime comparisons use paired medians from the same configuration. Where confidence intervals are reported, 95% intervals are obtained by bootstrap with configuration-level resampling for the frozen Planner evaluation. The Legal Oracle is an offline reference that selects the fastest measured candidate only from the legal candidate set available to TrustAero.

Table S1. Implementation and measurement environment.

Item	Frozen setting	Role
Implementation	Python 3.11; Pydantic 2.12.5; DuckDB 1.5.4	StrictParse → approval → planning → execution → lineage → Certificate → Checker
Machine	Windows 10; Intel Core i5-1135G7; 16 GB RAM; no GPU	Single-machine timing environment
Planner runs	DuckDB 1 thread; 1 GB memory limit	Warm-up, repeated measurements, rotated candidate order
Full-month execution	DuckDB 4 threads; 4 GB memory limit	BTS/NYC governed execution path
Statistics	Paired medians; 95% bootstrap CI; 3% practical-equivalence threshold	Held-out configurations are the resampling unit
Reproducibility	Fixed seeds; frozen protocols; registered outputs and manifests	Artifact paths are listed in Section J

Table S2. Datasets and experimental roles.

Dataset/workload	Scale	Primary role
Predefined cases	26 semantic cases; 11-case validation-ablation subset	Approval outcomes, major reasons, Certificate faults
Real-Agent	120 responses: 3 models × 2 modes × 20 tasks	Strict parsing and approval behavior
Plan mutations	4,000 unsafe + 1,000 legal controls; additional compound suites	UA/FR and repairability stress
BTS 2024	Mar/Jun/Sep/Dec; full-month input up to 547,271 rows	Governed-pipeline planning and source-level overhead
NYC TLC 2024	Mar/Jun/Sep/Dec; full-month input up to 2,964,624 rows	Cross-domain planning and source-level overhead
Synthetic	Selectivity/width/reuse sweeps; 100K/500K/1M lineage scales	Candidate diversity and record-level lineage
TPC-H SF1/SF10	Q1/Q3/Q6/Q10 plus governed join-aggregate family	Relational-query equivalence and third-family planning
BTS 2025	12 months; 7,001,619 records; 2,304 timed executions	Temporal holdout
Four-source	Earthquake, oil-well, airport, and city data	End-to-end approval/planning/execution/checking closure

B. Deterministic Plan Approval: Detailed Results

The main paper reports the compact approval summary. This section expands the Real-Agent, mutation, and independently generated black-box workloads while preserving the same approval semantics: ACCEPT and REWRITE produce an approved logical plan, whereas CLARIFY and REJECT terminate before physical planning.

Table S3. End-to-end approval effectiveness.

Evaluation	Task	n	Criterion	Result
Semantic	Predefined semantic cases	26	Status / major-reason match	26/26 (100%)
Real-Agent	Strict parsing	120	Parse success	119/120 (99.2%)
Real-Agent	Authorized-complete	48	Expected REWRITE	48/48 (100%)
Real-Agent	Insufficient-information	24	Expected CLARIFY	24/24 (100%)
Real-Agent	Unauthorized	24	Expected REJECT	24/24 (100%)
Real-Agent	Semantic attacks	24	Unsafe acceptance	0/24 (0%)
Mutation	Unsafe variants	4,000	Unsafe acceptance	0/4,000 (0%)
Mutation	Legal controls	1,000	False rejection	0/1,000 (0%)

Table S4. Validation-mechanism ablation (UA / FR counts).

Workload	n	No validation	Schema/Type	+Policy/Output	Full TrustAero
Predefined subset	11	9 / 0	2 / 0	0 / 1	0 / 0
Real-Agent	120	67 / 0	42 / 0	0 / 52	0 / 0
Plan mutations	5,000	4,000 / 0	616 / 0	0 / 491	0 / 0

UA = unsafe acceptance; FR = false rejection. The 11-case predefined subset is the frozen validation-ablation denominator.

Table S5. Additional compound-mutation stress suites.

Suite	Unsafe cases	Legal controls	UA	FR	Unhandled exceptions
Two-factor mutations	4,000	1,000	0	0	0
Three-factor mutations	572	200	0	0	0
Compound mutations across 11 plan shapes	2,541	1,100	0	0	0

B.1 Independent Black-Box Adversarial Evaluation

The standalone generator uses only the public CandidatePlan JSON Schema, one public ACCEPT seed, one public REWRITE seed, and a frozen list of 16 attack objectives. It imports only the Python standard library and cannot access TrustAero implementation code, diagnostics, tests, or prior mutation operators. The 1,000-case corpus was hashed before invoking the production Validator (fixed seed 20260830).

Table S6. Frozen black-box corpus and primary outcomes.

Metric	Result
Total cases	1,000
Adversarial cases	800 (50 per objective)
Valid controls	200
Unsafe acceptances	0/800 (0.0%)
False rejections	0/200 (0.0%)
Valid decision matches	200/200 (100.0%)
Unexpected exceptions	0/1,000 (0.0%)
Attack-objective coverage	16/16
Status distribution	750 REJECT; 50 CLARIFY; 100 ACCEPT; 100 REWRITE
Attacks passing StrictParse	600/800

Table S7. Results by registered black-box attack objective.

Objective	Cases	Safe outcome	Unsafe acceptance
Missing operator collection	50	50 REJECT	0
Unexpected plan property	50	50 REJECT	0
Unsupported schema version	50	50 REJECT	0
Unknown operator kind	50	50 REJECT	0
Duplicate operator identity	50	50 REJECT	0
Dangling input reference	50	50 REJECT	0
Cyclic dataflow	50	50 REJECT	0
Unknown declared output	50	50 REJECT	0
Unreachable subgraph	50	50 REJECT	0
Invalid operator arity	50	50 REJECT	0
Unknown dataset reference	50	50 REJECT	0
Unknown field reference	50	50 REJECT	0
Unavailable snapshot	50	50 REJECT	0
Missing declared purpose	50	50 CLARIFY	0
Unauthorized subject role	50	50 REJECT	0
Unknown requested output	50	50 REJECT	0

All ACCEPT and REWRITE outcomes in the 1,000-case corpus belong to valid controls. The supported claim is limited to these 16 registered objectives and the two public seed-plan families.

C. Held-out Physical-Planning Results

The frozen Planner evaluation contains 38 held-out configurations, each evaluated under three fixed seeds, for 114 decisions. The 96 governed-pipeline decisions come from 32 configurations and the 18 lineage-checkpoint decisions from 6 configurations. Seeds are nested within configurations; statistical analysis is performed at the configuration level.

Table S8. Frozen held-out Planner summary.

Candidate family	Method	Decisions	Within 3%	Mean regret	P95 regret	Max regret
Governed pipeline	TrustAero	96	100%	0.220%	1.442%	2.084%
Governed pipeline	Best Fixed Legal	96	75%	5.193%	20.443%	20.788%
Lineage checkpoint	TrustAero	18	100%	0.000%	0.000%	0.000%
Lineage checkpoint	Fixed-threshold baseline	18	83.3%	2.126%	13.510%	13.510%
Combined	TrustAero	114	100%	—	—	2.084%

Table S9. Ablation of governance-sensitive cost features.

Candidate family	Planner variant	n	Exact Oracle-set	Mean regret	Max regret
Governed pipeline	Full	48	100.00%	0.088%	2.120%
Governed pipeline	w/o Mask/Policy	48	18.75%	26.870%	81.170%
Governed pipeline	w/o Checkpoint	48	87.50%	0.985%	8.853%
Lineage checkpoint	Full	18	100.00%	0.000%	0.000%
Lineage checkpoint	w/o Lineage	18	72.22%	2.830%	13.510%

D. Hard Legality versus Soft Penalty Ranking

This experiment replays the same frozen 96-unit BTS/NYC workload under permissive, no-raw-join, and strict policies. Soft planning minimizes predicted_latency_ms + λ × violation_score, where the violation score is 0 for a legal candidate and 1 for a candidate violating at least one hard governance constraint. This is an offline replay and is not added to the original 114-decision denominator.

Table S10. Hard legality versus soft-penalty ranking.

Policy regime	Method	Illegal / 96	Within 3% / 96	Conditional mean / P95 / max regret
Permissive	Every λ and hard legality	0	96	0 / 0 / 0
No raw join	λ = 0, 0.1, or 1 ms	96	0	n/a
No raw join	λ = 10 ms	45	51	0.037 / 0.270 / 0.962%
No raw join	λ = 100 or 1,000 ms	0	96	0.220 / 1.442 / 2.084%
No raw join	Hard legality-first	0	96	0.220 / 1.442 / 2.084%
Strict	λ = 0, 0.1, or 1 ms	96	0	n/a
Strict	λ = 10 ms	45	51	0 / 0 / 0
Strict	λ = 100 or 1,000 ms	0	96	0 / 0 / 0
Strict	Hard legality-first	0	96	0 / 0 / 0

Table S11. Penalty magnitude required to eliminate the cheapest illegal winner.

Policy	Minimum	Median	P95	Maximum
No raw join	2.844 ms	7.408 ms	25.484 ms	26.417 ms
Strict	2.844 ms	7.408 ms	50.328 ms	51.646 ms

A 100 ms penalty reproduces hard pruning on this bounded workload, but only after dominating the observed workload- and policy-dependent cost gaps. Hard legality-first enforces the same feasibility boundary without penalty calibration.

E. Third Governed Join-Aggregate Candidate Family

The third physical-candidate family contains three registered legal implementations of the same governed join-aggregate: governed-fact-first, eligible-dimension-first, and partial-aggregate-first. All candidates apply the same policy filter and deterministic masking, produce identical canonical results, and have distinct observed DuckDB physical fingerprints.

Table S12. Development admission and frozen SF10 holdout.

Stage	Scale	Key result
SF1 development	8 configurations × 3 salts = 24 units; 432 timings	5/8 singleton-winner configurations; eligible-dimension-first wins 3, governed-fact-first 1, partial-aggregate-first 1; selector frozen before SF10
Primary SF10 holdout	4 unseen configurations × 3 new salts = 12 decisions; 216 timings	11/12 within 3%; mean regret 0.614%; max regret 4.463%; 0 illegal; all result-equivalence and distinct-plan gates pass
Exact-protocol stability replication	Same 12 decisions; 216 timings	12/12 within 3%; mean regret 0.187%; max regret 2.239%; 0 illegal; reported separately from the primary run

The sole primary decision above 3% is p0.400-q0.600-s6203. The frozen selector chose eligible-dimension-first at 14,320.5 ms; governed-fact-first measured 13,708.7 ms, yielding 4.463% regret. Scenario-level hierarchical paired-bootstrap analysis places both candidates in the confidence-undominated set. In the exact-protocol replication, eligible-dimension-first measured 12,720.0 ms and governed-fact-first 13,537.4 ms. The primary one-shot result remains the held-out result reported by the paper.

F. Candidate-Space Planning Scalability

The scalability study composes five genuine decision axes: three execution orders, two materialization boundaries, two lineage-checkpoint placements, two evidence-capture modes, and two Mask placements. This yields nested spaces of 3, 6, 12, 24, and 48 structurally distinct candidates with unique physical fingerprints. The study covers all 96 frozen BTS/NYC configurations, three governance regimes, five warmups, and 50 measured repetitions, for 72,000 planning trials.

Table S13. Candidate-space planning latency (microseconds).

Candidates	Policy	Legal	Generation	Filter	Ranking	Total median	Total P95
3	Permissive	3	24.3	0.9	0.9	26.1	32.9
6	Permissive	6	47.0	1.3	1.2	49.6	59.1
12	Permissive	12	93.6	2.1	1.8	97.5	130.4
24	Permissive	24	184.9	3.5	2.8	191.2	225.1
48	Permissive	48	367.1	6.9	4.9	379.0	545.2
48	No raw join	24	368.0	6.4	2.9	377.2	536.3
48	Strict	6	369.5	6.8	1.4	377.7	582.0

Across all 15 scale-policy groups, 72,000/72,000 trials completed with zero illegal selections and exact registered cost-oracle agreement. This is a control-plane scalability result, not a new physical-runtime or general SQL-optimization claim.

G. Temporal and Conventional Relational Coverage

G.1 BTS 2025 Temporal Holdout

Table S14. Full-year temporal holdout of physical-plan candidates on BTS 2025.

Physical realization	Materialization boundary	Unique-winner months	Median monthly runtime ratio	Monthly ratio range
Fused	None	12/12	1.00×	1.00–1.00×
Materialize after filtering	After flight filtering	0/12	1.49×	1.47–1.70×
Materialize after origin join	After origin-airport join	0/12	1.77×	1.60–1.97×
Materialize after carrier join	After carrier join	0/12	1.98×	1.74–2.10×

The temporal holdout contains 12 monthly snapshots, 7,001,619 input records, and 2,304 timed candidate executions. All four realizations produced equivalent results in every month. BTS 2025 was not used for candidate design, estimator fitting, or threshold selection and is not included in the 114-decision frozen Planner evaluation.

G.2 Conventional TPC-H Query Coverage

For the four TPC-H queries supported by the current typed IR (Q1, Q3, Q6, and Q10), TrustAero reproduces the corresponding official SQL results exactly at both SF1 and SF10. Equivalent physical refinements exhibit query-dependent runtime diversity: at SF10, the fastest and slowest registered candidates differ by approximately 9.29× for Q1, 5.43× for Q6, and 1.02× for Q10. Q3 is included in the result-equivalence coverage; its runtime-diversity ratio is not used as a headline result in the current paper.

H. Certificate, Checker, and End-to-End Evidence

Table S15. Certificate evidence and Checker robustness.

Evaluation	Configuration	Cases	Correct outcomes	Rate
Certificate evidence	Event only	19	6/19	31.6%
Certificate evidence	Event + Lineage	19	8/19	42.1%
Certificate evidence	Full Certificate	19	19/19	100.0%
Component removal	w/o Planner decision	19	13/19	68.4%
Component removal	w/o Events	19	14/19	73.7%
Component removal	w/o Snapshot	19	16/19	84.2%
Component removal	w/o Result	19	17/19	89.5%
Component removal	w/o Lineage	19	17/19	89.5%

Evaluation	Configuration	Cases	Correct outcomes	Rate
Checker robustness	Pairwise tampering	10	10/10	100.0%
Checker robustness	Combinatorial tampering	92	92/92	100.0%
Checker robustness	Invalid Event DAGs	1,000	1,000/1,000	100.0%
Checker robustness	Cross-fixture invalid cases	24,000	24,000/24,000	100.0%
Checker robustness	Cross-fixture valid controls	2,002	2,002/2,002	100.0%

H.1 Four-Source End-to-End Closure

Table S16. Four-source governed execution path.

Stage	Observed outcome
Logical approval	Safe request returns REWRITE; deterministic governance adds Mask, location governance, and source-level lineage obligations
Physical planning	4 candidates generated; 3 removed for sensitive-value exposure or materialization violations; 1 legal 13-operator plan approved
Execution	9,128 result rows; zero raw-sensitive-value exposure; four pinned data snapshots
Evidence	Source-level lineage binds all four frozen snapshots; untampered Certificate returns PARTIAL
Directed tampering	Result, policy, snapshot, Planner decision, lineage, and Event-dependency modifications are all rejected by the Checker

This case exercises the complete chain from logical rewriting and legal candidate filtering to governed execution, Certificate construction, and independent checking in one multi-source spatial-analysis workflow.

I. Lineage Runtime and Storage Details

Table S17. Governed execution and lineage overhead.

Configuration	Runtime / Direct SQL	95% CI	Checker / stage latency	Storage / reduction
BTS source-level	0.969×	[0.902, 1.049]	Checker 0.124 ms	Source-level
NYC TLC source-level	1.040×	[0.986, 1.058]	Checker 0.131 ms	Source-level
Row-ordinal 100K	2.117×	[2.035, 2.191]	—	~32 B/edge; 49.5% latency reduction vs key-based
Row-ordinal 500K	1.935×	[1.876, 1.982]	Capture 2.93 s; Checker 2.68 s	~32 B/edge; 62.8% latency reduction vs key-based
Row-ordinal 1M	2.045×	[2.015, 2.145]	—	32.000462 B/edge

Key-based record-lineage encoding stores approximately 64 B per edge; row-ordinal encoding derives the output endpoint from the canonicalized result digest and row ordinal, reducing identifier storage to approximately 32 B per edge. The current record-level lineage path is evaluated separately from the four-source source-level lineage case.

J. Artifact Organization and Reproduction

The public artifact repository is available at https://github.com/yunguinie/TrustAero.

The repository contains the implementation, frozen experiment configurations, registered workloads, result files, and scripts used to reproduce the results reported in the main paper and this supplementary material. Experiment-specific paths and reproduction commands are documented in the repository README.